# The genus-2 free energy of $W_3$ and the gravitational coproduct

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*Remark* 0.1 (Relationship to companion papers). The $W_3$ composite field identity $\Lambda_0|h\rangle = h^2 - 9h/5$ and the complete rank-2 holographic datum are developed in the companion paper *The first rank-2 holographic datum: $\mathcal{H}(W_3)$* (`w3_holographic_datum.tex`).

## 1 The genus-2 free energy of $W_3$

The preceding sections treated the single-channel regime: one strong generator, one primary line, one modular characteristic $\kappa$. We now compute the first multi-channel invariant. The $W_3$ algebra has two strong generators (the stress tensor $T$ of weight 2 and the spin-3 current $W$ of weight 3), and $F_2(W_3)$ receives contributions from both channels and their interactions.

### 1.1 The Frobenius algebra of the bar-complex CohFT

The bar-complex CohFT of $W_3$ has a two-dimensional Frobenius algebra with basis $\{e_T, e_W\}$ and diagonal metric from the per-channel modular characteristics:

$$\eta_{TT} = \kappa_T = \frac{c}{2}, \qquad \eta_{WW} = \kappa_W = \frac{c}{3}, \qquad \eta_{TW} = 0. \tag{1}$$

The total modular characteristic is $\kappa(W_3) = \kappa_T + \kappa_W = 5c/6 = c \cdot (H_3 - 1)$, where $H_3 = 1 + 1/2 + 1/3 = 11/6$ is the third harmonic number. The inverse metric gives the propagators:

$$\eta^{TT} = \frac{2}{c}, \qquad \eta^{WW} = \frac{3}{c}. \tag{2}$$

Both propagators have weight 1: the bar-complex propagator is $d\log E(z,w)$, which has weight 1 regardless of the conformal weight of the field being sewed.

The OPE determines the three-point structure constants. The $\mathbb{Z}_2$ symmetry $W \mapsto -W$ kills all three-point functions with an odd number of $W$-insertions:

$$C_{TTT} = C_{TWW} = C_{WWT} = c, \qquad C_{TTW} = C_{TWT} = C_{WTT} = C_{WWW} = 0. \tag{3}$$

The derivations are: $T_{(1)}T = 2T$ gives $C^T_{TT} = 2$, whence $C_{TTT} = \eta_{TT} \cdot 2 = c$; $T_{(1)}W = 3W$ gives $C^W_{TW} = 3$, whence $C_{TWW} = \eta_{WW} \cdot 3 = c$; $W_{(3)}W = 2T$ gives $C^T_{WW} = 2$, whence $C_{WWT} = \eta_{TT} \cdot 2 = c$.

The genus-0 four-point vertex is universal: for any channel pair $(i,j) \in \{T,W\}^2$,

$$V_{0,4}(i,i \mid j,j) \;=\; \sum_m \eta^{mm}\, C_{iim}\, C_{jjm} \;=\; \eta^{TT} \cdot c \cdot c \;=\; 2c. \tag{4}$$

Only the $T$-channel contributes to the sum: $C_{iiT} = c$ for $i \in \{T,W\}$, while $C_{iiW} = 0$ for both $i = T$ (since $C_{TTW} = 0$) and $i = W$ (since $C_{WWW} = 0$). The $W$-channel is invisible at genus-0 valence 4.

## 1.2 The seven stable graphs at genus 2

The moduli space $\overline{\mathcal{M}}_{2,0}$ has a stratification by seven stable graphs $\Gamma_0, \ldots, \Gamma_6$:

| | **Name** | **Vertices** | **Edges** | $\lvert\mathrm{Aut}\rvert$ | $h^1$ | **Type** |
|---|---|---|---|---|---|---|
| $\Gamma_0$ | smooth | ($g$=2, val=0) | 0 | 1 | 0 | compact stratum |
| $\Gamma_1$ | figure-eight | ($g$=1, val=2) | 1 self-loop | 2 | 1 | one-loop |
| $\Gamma_2$ | banana | ($g$=0, val=4) | 2 self-loops | 8 | 2 | two-loop sunset |
| $\Gamma_3$ | dumbbell | ($g$=1, val=1) + ($g$=1, val=1) | 1 bridge | 2 | 0 | separating |
| $\Gamma_4$ | theta | ($g$=0, val=3) + ($g$=0, val=3) | 3 bridges | 12 | 2 | non-separating |
| $\Gamma_5$ | tadpole | ($g$=0, val=3) + ($g$=1, val=1) | 1 self-loop + 1 bridge | 2 | 1 | mixed |
| $\Gamma_6$ | chain | ($g$=0, val=3) + ($g$=0, val=3) | 2 self-loops + 1 bridge | 8 | 2 | non-separating |

Every vertex satisfies the stability condition $2g_v + \mathrm{val}_v \geq 3$, and the genus formula $h^1(\Gamma) + \sum_v g_v = 2$ holds in each case. Note that the smooth graph $\Gamma_0$ has no edges and contributes the integrated class on $\mathcal{M}_{2,0}$ itself; the remaining six graphs are the boundary strata.

## 1.3 Multi-channel Feynman rules

For each graph $\Gamma$ with $|E(\Gamma)|$ edges, a *channel assignment* is a function $\sigma\colon E(\Gamma) \to \{T, W\}$. The Feynman rules of the bar-complex CohFT assign to each pair $(\Gamma, \sigma)$ an amplitude

$$\mathcal{A}(\Gamma, \sigma) \;=\; \frac{1}{|\mathrm{Aut}(\Gamma)|} \prod_{e \in E(\Gamma)} \eta^{\sigma(e),\,\sigma(e)} \prod_{v \in V(\Gamma)} V_v\big(g_v;\, \sigma|_{\text{half-edges}(v)}\big), \tag{5}$$

where the vertex factors are:

- *Genus-0, trivalent*: $V_v = C_{\sigma(e_1)\sigma(e_2)\sigma(e_3)}$.
- *Genus-0, four-valent* (the banana vertex): $V_v = \sum_m \eta^{mm} C_{\sigma(e_1)\sigma(e_2)m}\, C_{m\,\sigma(e_3)\sigma(e_4)}$.
- *Genus-1, valence $n$*: $V_v = \kappa_{\sigma(e)}/24$ (per-channel genus-1 universality, proved unconditionally).

Each half-edge at a self-loop contributes a pair of half-edges to the same vertex, both carrying channel $\sigma(e)$.

## 1.4 Per-graph amplitudes

We compute each boundary graph amplitude, decomposed into *diagonal* (all edges in one channel) and *mixed* (edges in different channels) contributions.

### $\Gamma_1$: figure-eight

One edge, two channel assignments. The vertex is genus-1 with valence 2, so the vertex factor for channel $i$ is $\kappa_i/24$:

$$\mathcal{A}(\Gamma_1, T) = \tfrac{1}{2}\,\eta^{TT} \cdot \tfrac{\kappa_T}{24} = \tfrac{1}{2} \cdot \tfrac{2}{c} \cdot \tfrac{c}{48} = \tfrac{1}{48}, \tag{6}$$

$$\mathcal{A}(\Gamma_1, W) = \tfrac{1}{2}\,\eta^{WW} \cdot \tfrac{\kappa_W}{24} = \tfrac{1}{2} \cdot \tfrac{3}{c} \cdot \tfrac{c}{72} = \tfrac{1}{48}. \tag{7}$$

Both channels contribute equally: the propagator $1/\kappa_i$ cancels the genus-1 vertex factor $\kappa_i/24$, leaving the universal value $1/48$ independent of $c$. The total is $1/24 = \lambda_1^{\mathrm{FP}}$. No mixed channel assignments exist (single edge).

### $\Gamma_2$: banana

Two edges, four channel assignments, $|\mathrm{Aut}| = 8$. The vertex is genus-0 with valence 4, and the four-point vertex factor is $V_{0,4}(i, i \mid j, j) = 2c$ universally (4).

$$\mathcal{A}(\Gamma_2, TT) = \tfrac{1}{8}\,(\eta^{TT})^2 \cdot 2c = \tfrac{1}{8} \cdot \tfrac{4}{c^2} \cdot 2c = \frac{1}{c}, \tag{8}$$

$$\mathcal{A}(\Gamma_2, WW) = \tfrac{1}{8}\,(\eta^{WW})^2 \cdot 2c = \tfrac{1}{8} \cdot \tfrac{9}{c^2} \cdot 2c = \frac{9}{4c}, \tag{9}$$

$$\mathcal{A}(\Gamma_2, TW{+}WT) = \tfrac{1}{8} \cdot 2 \cdot \eta^{TT}\,\eta^{WW} \cdot 2c = \tfrac{1}{8} \cdot 2 \cdot \tfrac{6}{c^2} \cdot 2c = \frac{3}{c}. \tag{10}$$

The cross-channel contribution is $\delta_{\mathrm{banana}} = 3/c$.

### $\Gamma_3$: dumbbell

One bridge edge, two channel assignments, $|\mathrm{Aut}| = 2$. Each vertex has genus 1 and valence 1, with vertex factor $\kappa_i/24$:

$$\mathcal{A}(\Gamma_3, T) = \tfrac{1}{2}\,\eta^{TT} \cdot \left(\tfrac{\kappa_T}{24}\right)^2 = \tfrac{1}{2} \cdot \tfrac{2}{c} \cdot \tfrac{c^2}{2304} = \frac{c}{2304}, \tag{11}$$

$$\mathcal{A}(\Gamma_3, W) = \tfrac{1}{2}\,\eta^{WW} \cdot \left(\tfrac{\kappa_W}{24}\right)^2 = \tfrac{1}{2} \cdot \tfrac{3}{c} \cdot \tfrac{c^2}{5184} = \frac{c}{3456}. \tag{12}$$

The total is $c/2304 + c/3456 = (3c + 2c)/6912 = 5c/6912 = \kappa(W_3)/1152$. No mixed channel assignments exist (single edge).

### $\Gamma_4$: theta

Three bridge edges, eight channel assignments, $|\mathrm{Aut}| = 12$. Each vertex has genus 0 and valence 3, with vertex factor $C_{\sigma(e_1)\sigma(e_2)\sigma(e_3)}$.

- *All-T*: both vertices get $C_{TTT} = c$. Amplitude $= \frac{1}{12}\,(\eta^{TT})^3 \cdot c^2 = \frac{1}{12} \cdot \frac{8}{c^3} \cdot c^2 = \frac{2}{3c}$.
- *All-W*: both vertices get $C_{WWW} = 0$. Amplitude $= 0$.
- *Type* $(T, T, W)$ (three such assignments): each vertex receives one $W$ half-edge, giving $C_{TTW} = 0$. All three vanish.

- *Type* $(T, W, W)$ (three such assignments): each vertex receives half-edges $(T, W, W)$, giving $C_{TWW} = c$. The propagator product is $\eta^{TT}(\eta^{WW})^2 = (2/c)(9/c^2) = 18/c^3$. Each assignment contributes $18c^2/c^3 = 18/c$; with $|\mathrm{Aut}| = 12$, the total mixed amplitude is

$$\delta_{\text{theta}} = \frac{3 \cdot 18}{12c} = \frac{9}{2c}. \tag{13}$$

**$\Gamma_5$: tadpole**

One self-loop at vertex $v_0$ (genus 0, valence 3) and one bridge to vertex $v_1$ (genus 1, valence 1). Two edges, four channel assignments, $|\mathrm{Aut}| = 2$.

- $(T, T)$: self-loop $T$, bridge $T$. Vertex $v_0$ gets half-edges $(T, T, T) \to C_{TTT} = c$. Vertex $v_1$ gets half-edge $T \to \kappa_T/24 = c/48$. Propagators: $(\eta^{TT})^2 = 4/c^2$. With $|\mathrm{Aut}|$: $\frac{1}{2} \cdot \frac{4}{c^2} \cdot c \cdot \frac{c}{48} = \frac{1}{24}$.
- $(W, W)$: self-loop $W$, bridge $W$. Vertex $v_0$ gets $(W, W, W) \to C_{WWW} = 0$. Vanishes.
- $(T, W)$: self-loop $T$, bridge $W$. Vertex $v_0$ gets $(T, T, W) \to C_{TTW} = 0$. Vanishes.
- $(W, T)$: self-loop $W$, bridge $T$. Vertex $v_0$ gets $(W, W, T) \to C_{WWT} = c$. Vertex $v_1$ gets $T \to \kappa_T/24 = c/48$. Propagators: $\eta^{WW} \cdot \eta^{TT} = (3/c)(2/c) = 6/c^2$. With $|\mathrm{Aut}|$:

$$\delta_{\text{tadpole}} = \tfrac{1}{2} \cdot \tfrac{6}{c^2} \cdot c \cdot \tfrac{c}{48} = \tfrac{6c^2}{96c^2} = \frac{1}{16}. \tag{14}$$

The tadpole mixed amplitude $1/16$ is *independent of* $c$: the $c^2$ in the numerator (from $C_{WWT} \cdot \kappa_T/24$) cancels the $c^2$ in the denominator (from $\eta^{WW} \cdot \eta^{TT}$).

**$\Gamma_6$: chain (barbell)**

Two genus-0 trivalent vertices, each carrying one self-loop, connected by a bridge. Three edges (self-loop $s_1$, self-loop $s_2$, bridge $b$), eight channel assignments, $|\mathrm{Aut}| = 8$. Each vertex $v_j$ receives half-edges $(\sigma(s_j), \sigma(s_j), \sigma(b))$ with vertex factor $C_{\sigma(s_j)\,\sigma(s_j)\,\sigma(b)}$.

Only $\sigma(b) = T$ gives nonzero vertex factors (since $C_{TTW} = C_{WWW} = 0$). Among the four remaining assignments $(\sigma(s_1), \sigma(s_2)) \in \{T, W\}^2$ with $\sigma(b) = T$:

- $(T, T, T)$: diagonal. Vertex factors $C_{TTT}^2 = c^2$, propagators $(\eta^{TT})^3 = 8/c^3$. With $|\mathrm{Aut}|$: $(1/8)(8/c^3)(c^2) = 1/c$.
- $(T, W, T)$: mixed. $C_{TTT} \cdot C_{WWT} = c^2$, $\eta^{TT}\,\eta^{WW}\,\eta^{TT} = 12/c^3$. With $|\mathrm{Aut}|$: $(1/8)(12/c^3)(c^2) = 3/(2c)$.
- $(W, T, T)$: mixed. By the $s_1 \leftrightarrow s_2$ symmetry: $3/(2c)$.
- $(W, W, T)$: mixed. $C_{WWT}^2 = c^2$, $(\eta^{WW})^2\,\eta^{TT} = 18/c^3$. With $|\mathrm{Aut}|$: $(1/8)(18/c^3)(c^2) = 9/(4c)$.

The total mixed amplitude is

$$\delta_{\text{barbell}} = \frac{3}{2c} + \frac{3}{2c} + \frac{9}{4c} = \frac{6+6+9}{4c} = \frac{21}{4c}. \tag{15}$$

### 1.5 The cross-channel correction

Collecting the mixed-channel amplitudes from all boundary graphs:

$$\delta F_2 = \underbrace{\frac{3}{c}}_{\text{banana}} + \underbrace{\frac{9}{2c}}_{\text{theta}} + \underbrace{\frac{1}{16}}_{\text{tadpole}} + \underbrace{\frac{21}{4c}}_{\text{barbell}} = \frac{51}{4c} + \frac{1}{16} = \frac{c+204}{16c}. \tag{16}$$

The figure-eight and dumbbell graphs contribute zero mixed amplitude (single edges admit only diagonal assignments). The correction decomposes into two structurally distinct pieces:

- A *c-dependent* piece $51/(4c)$, arising from genus-0 vertex factors on the banana ($3/c$), theta ($9/(2c)$), and barbell ($21/(4c)$) graphs. This is $R$-matrix independent: genus-0 vertices receive no corrections from the CohFT $R$-matrix.
- A *c-independent* piece $1/16$, arising from the tadpole graph, which mixes a genus-0 vertex factor with a genus-1 vertex factor $\kappa_i/24$.

*Remark* 1.1 (Comparison with the scalar approximation). The per-channel (diagonal) graph sum, computed by applying Theorem D to each channel independently, gives

$$F_2^{\text{scal}}(W_3) = \kappa \cdot \lambda_2^{\text{FP}} = \frac{5c}{6} \cdot \frac{7}{5760} = \frac{7c}{6912}. \tag{17}$$

The ratio of the naive cross-channel correction to the scalar approximation is

$$\frac{\delta F_2}{F_2^{\text{scal}}} = \frac{432(c+204)}{7c^2}, \tag{18}$$

which diverges as $c \to 0$ and decays as $c^{-1}$ at large $c$. At $c = 4$: the ratio is approximately 802, and the cross-channel correction *dominates* the scalar term. At $c = 50$: the ratio is approximately 6.3. At large $c$, the tadpole's $c$-independent contribution $1/16$ dominates, and the correction becomes a subleading $O(1/c)$ effect relative to the $O(c)$ scalar term.

*Remark* 1.2 (Status of the cross-channel correction). The computation above uses the naive genus-1 vertex factors $\kappa_i/24$ without $R$-matrix corrections. The multi-generator universality problem (**op:multi-generator-universality**) is resolved negatively: the CohFT $R$-matrix does *not* produce compensating terms that cancel $\delta F_2$. The cross-channel correction $\delta F_2 = (c{+}204)/(16c)$ is a genuine additional term, and the correct decomposition is $F_g = \kappa \cdot \lambda_g^{\text{FP}} + \delta F_g^{\text{cross}}$.

### 1.6 The complete amplitude table

We record the full per-graph data:

| **Graph** | **Diagonal ($T$)** | **Diagonal ($W$)** | **Mixed** | **Total** |
|---|---|---|---|---|
| $\Gamma_1$ figure-eight | $1/48$ | $1/48$ | $0$ | $1/24$ |
| $\Gamma_2$ banana | $1/c$ | $9/(4c)$ | $3/c$ | $25/(4c)$ |
| $\Gamma_3$ dumbbell | $c/2304$ | $c/3456$ | $0$ | $5c/6912$ |
| $\Gamma_4$ theta | $2/(3c)$ | $0$ | $9/(2c)$ | $31/(6c)$ |
| $\Gamma_5$ tadpole | $1/24$ | $0$ | $1/16$ | $5/48$ |
| $\Gamma_6$ barbell | $1/c$ | $0$ | $21/(4c)$ | $25/(4c)$ |

The table records the six boundary-graph amplitudes; the smooth graph $\Gamma_0$ has no edges and contributes an integrated class on $\mathcal{M}_{2,0}$ itself. The mixed column sums to $\delta F_2 = (c+204)/(16c)$.

Every entry in this table has been verified against the compute module `w3_genus2.py`, using exact rational arithmetic (`fractions.Fraction`), at the seven test central charges $c \in \{1, 2, 4, 13, 26, 50, 100\}$.

### 1.7 Planted-forest corrections

The planted-forest correction on each primary line is $\delta_{\mathrm{pf}}^{(2,0)} = S_3(10S_3 - \kappa)/48$.

*T-line*: $S_3^T = 2$ (the Virasoro cubic shadow), $\kappa_T = c/2$.

$$\delta_{\mathrm{pf}}^T = \frac{2(20 - c/2)}{48} = \frac{40 - c}{48}. \tag{19}$$

This vanishes at $c = 40$, is positive for $c < 40$, and negative for $c > 40$.

*W-line*: $S_3^W = 0$ by the $\mathbb{Z}_2$ parity $W \mapsto -W$, which forces all odd-degree shadows on the $W$-line to vanish. Hence $\delta_{\mathrm{pf}}^W = 0$.

The total planted-forest correction is $\delta_{\mathrm{pf}} = (40 - c)/48$, living entirely on the $T$-line.

### 1.8 Koszul duality constraints

Under the Koszul involution, $W_3$ at central charge $c$ is dual to $W_3$ at central charge $c' = 100 - c$ (the Feigin–Frenkel involution for $\mathfrak{sl}_3$). The modular characteristics satisfy

$$\kappa(c) + \kappa(100 - c) = \frac{5c}{6} + \frac{5(100 - c)}{6} = \frac{250}{3}, \tag{20}$$

and the scalar free energies satisfy $F_2^{\mathrm{scal}}(c) + F_2^{\mathrm{scal}}(100 - c) = (250/3) \cdot \lambda_2^{\mathrm{FP}}$. The cross-channel correction at the dual pair is

$$\delta F_2(c) + \delta F_2(100 - c) = \frac{c + 204}{16c} + \frac{304 - c}{16(100 - c)} = -\frac{c^2 - 100c - 10200}{8c(100 - c)}.$$

This sum is not constant in $c$: the cross-channel correction does not respect Koszul additivity. If universality holds ($\delta F_2$ is cancelled by $R$-matrix corrections), this is automatic. If the naive correction persists, it produces a Koszul-asymmetric contribution to $F_2$: an observable departure from single-channel behaviour.

## 2 The gravitational coproduct is primitive

In gauge theory, gluons split: one gluon can exit a scattering region as two, and the dg-shifted Yangian coproduct encodes this splitting. In gravity, the graviton is composite, not fundamental. The stress tensor is the Sugawara quadratic

$$T = \frac{1}{k+2}\Big(\tfrac{1}{2}{:}J^0 J^0{:} + {:}J^+ J^-{:}\Big), \tag{21}$$

built from affine currents by normal ordering. This compositeness has a precise algebraic consequence: the transferred coproduct is trivial.

## 2.1 The DS-HPL transfer theorem

The Drinfeld–Sokolov reduction $\hat{\mathfrak{sl}}_2 \to \mathrm{Vir}$ is a BRST reduction with a strong deformation retract $(i, p, h)$ from the BRST complex to the Virasoro cohomology. The homotopy $h$ has ghost-number degree $-1$: it maps $V \mapsto b$ (ghost number $0 \to -1$) and $c \mapsto U/2$ (ghost number $1 \to 0$).

**Theorem 2.1** (DS-HPL transfer)**.** *Let $Y^{\mathrm{dg}}(\widehat{\mathfrak{sl}}_2)$ denote the dg-shifted Yangian at level $k$, with $A_\infty$ products $\{m_n^{\mathrm{aff}}\}$, coproduct $\Delta_z^{\mathrm{aff}}$, and $r$-matrix $r^{\mathrm{aff}}(z) = \Omega/((k+2)z)$. The HPL transfer through the BRST SDR produces:*

(i) *Transferred $A_\infty$ products $\{m_n^{\mathrm{Vir}}\}_{n\geq 2}$ on the Virasoro sector, with $m_n^{\mathrm{Vir}} \neq 0$ for all $n \geq 2$ (the entire infinite $A_\infty$ tower of class $\mathbf{M}$).*

(ii) *A transferred coproduct $\Delta_z^{\mathrm{Vir}} \colon H_{\mathrm{Vir}} \to H_{\mathrm{Vir}} \otimes H_{\mathrm{Vir}}$.*

(iii) *A transferred $r$-matrix $r^{\mathrm{Vir}}(z) = (c/2)/z^3 + 2T/z$, with cubic leading pole reflecting the Sugawara nonlinearity.*

The affine $r$-matrix $r^{\mathrm{aff}}(z) = \Omega/((k+2)z)$ has a simple pole (the OPE has at most a double pole; the bar kernel $d\log(z-w)$ absorbs one power). The Virasoro OPE has a quartic pole ($z^{-4}$ in $T(z)T(w)$); absorption produces the cubic pole $z^{-3}$ of $r^{\mathrm{Vir}}(z)$. The general principle: $r$-matrix pole order = OPE pole order minus one.

**Theorem 2.2** (Gravitational coproduct primitivity)**.** *The HPL-transferred coproduct is strictly primitive at all degrees:*

$$\Delta_{z,k}^{\mathrm{Vir}} = 0 \qquad \textit{for all } k \geq 2. \tag{22}$$

*That is, $\Delta_z^{\mathrm{Vir}}(T) = \tau_z(T) \otimes 1 + 1 \otimes T$. All nontrivial structure of the Virasoro dg-shifted Yangian resides in the $r(z)$-twisted target product, never in the coproduct.*

*Proof.* The proof is a ghost-number obstruction, proceeding in three steps.

*Step 1.* The linear transferred coproduct $(p \otimes p)(\Delta_z(T))$ is already primitive. The affine coproduct $\Delta_z(T)$ expands as $\tau_z(T) \otimes 1 + 1 \otimes T + \frac{1}{k+2}\,\Omega(w+z,\, w)$, where $\Omega(w+z, w) = J^0(w+z) \otimes J^0(w) + J^+(w+z) \otimes J^-(w) + J^-(w+z) \otimes J^+(w)$. The projection $p$ annihilates $J^0$ (not in the Virasoro sector) and $J^+$ decomposes as $V + 1$ with $p(V) = 0$. Every summand of $\Omega$ contains at least one factor from $\{J^0, J^-, J^+\}$ at conformal weight 1, where the Virasoro sector has no states ($L_{-1}|0\rangle = 0$). Each tensor factor of $(p \otimes p)(\Omega)$ has $p$ killing at least one component, so $(p \otimes p)(\Omega) = 0$.

*Step 2.* At degree $k \geq 2$, every HPL tree has at least one internal edge labelled by the homotopy $h$. The ghost-number accounting is:

| Operation | Ghost degree |
|---|---|
| $h$ (homotopy) | $-1$ |
| $\delta$ (BRST perturbation) | $+1$ |
| $\mu$ (vertex algebra product) | 0 |
| $\Delta_z$ (coproduct) | 0 |

The product $h\delta$ has ghost degree $-1 + 1 = 0$, but the tree topology at degree $k \geq 2$ always contains one *uncompensated* $\delta$: a binary tree with $k$ leaves has $k-1$ internal vertices (each carrying $\delta$, ghost $+1$) and $k-2$ internal edges (each carrying $h$, ghost $-1$), giving total ghost number $(k-1) - (k-2) = +1$. The overall output before projection has ghost number $+1$.

*Step 3.* The W-algebra cohomology sits in ghost number 0. The projection $p$ annihilates all ghost-number-($\neq 0$) elements. Since the output of each degree-$k$ tree has ghost number $+1$, the projection $(p \otimes p)$ kills every summand.

The perturbation does not produce corrections. The BRST perturbation $\delta$ has ghost degree $+1$, so the product $h\delta$ preserves ghost number. But $\delta(T) = 0$ (the stress tensor is BRST-closed), so $(h\delta)(i(T)) = 0$, and by induction $(h\delta)^n(i(T)) = 0$ for all $n \geq 1$. The corrected inclusion $i_\infty(T) = i(T)$ is unperturbed, and the same primitive coproduct results at every perturbative order. □

**Conjecture 2.3** (General DS primitivity)**.** *For any simple Lie algebra $\mathfrak{g}$, the HPL-transferred coproduct of the principal $\mathcal{W}$-algebra $\mathcal{W}(\mathfrak{g})$ is strictly primitive at all degrees. The ghost-number obstruction of the BRST complex for the principal Drinfeld–Sokolov reduction $V_k(\mathfrak{g}) \to \mathcal{W}_k(\mathfrak{g})$ forces $\Delta^{\mathcal{W}}_{z,n} = 0$ for $n \geq 2$.*

## 2.2 Physical meaning

The coproduct $\Delta_z$ of a dg-shifted Yangian encodes the *splitting* of excitations. Its strictness or nontriviality is the algebraic fingerprint of the distinction between fundamental and composite fields.

| **Theory** | **Boundary algebra** | $\boldsymbol{\Delta_z}$ | **Shadow depth** | **Reason** |
|---|---|---|---|---|
| Chern–Simons | $\hat{\mathfrak{g}}_k$ (affine KM) | $\neq \varepsilon$ | 3 (class **L**) | fundamental gluons split |
| 3d gravity | $\mathrm{Vir}_c = W_k(\mathfrak{sl}_2)$ | $= \varepsilon$ | $\infty$ (class **M**) | ghost obstruction (DS) |
| Twisted holography | $H_k$ (Heisenberg) | $= \varepsilon$ | 2 (class **G**) | free field, $r = k/z$ |
| M2 brane | DDCA | $\neq \varepsilon$ | — | no DS, fundamental modes |
| M5 brane | $W_{1+\infty}(K)$ | $= \varepsilon$ | $\infty$ (class **M**) | ghost obstruction (DS) |

The pattern: *principal Drinfeld–Sokolov reductions produce primitive coproducts.* The BRST ghost sector provides the grading obstruction that kills all higher-degree components. Theorem 2.2 proves this for $\widehat{\mathfrak{sl}}_2 \to \mathrm{Vir}$; the same argument applies to any principal DS reduction $\widehat{\mathfrak{g}} \to \mathcal{W}(\mathfrak{g})$ (the ghost-number grading is universal), yielding Conjecture 2.3. Theories with fundamental excitations (Chern–Simons, deformed double current algebras) lack a ghost sector, and their coproducts are nontrivial.

The two exceptions to the DS pattern are illuminating. The Heisenberg algebra has a primitive coproduct for a different reason: it is a free field, its $r$-matrix $r^H(z) = k/z$ is scalar, and the bar complex is formal at degree 2 (shadow depth $r_{\max} = 2$, class **G**). No DS reduction and no ghost obstruction; primitivity follows from the absence of nonlinear OPE structure. The M2 brane boundary algebra (a deformed double current algebra) has a nontrivial coproduct because it is not obtained by DS reduction from any affine algebra; its excitations are fundamental modes of the M2 worldvolume theory.

## 2.3 The $r$-matrix carries all gravitational structure

With the coproduct primitive, the full gravitational content of the Virasoro dg-shifted Yangian is encoded in the $r$-matrix

$$r^{\mathrm{Vir}}(z) = \frac{c/2}{z^3} + \frac{2T}{z}. \tag{23}$$

The two terms encode distinct physical data:

- The *leading coefficient* $c/2 = \kappa(\mathrm{Vir}_c)$ is the modular characteristic. It determines the genus tower $F_g = (c/2)\int_{\overline{\mathcal{M}}_g} \lambda_g$ (uniform-weight) at all genera, the BTZ black hole entropy $S_{\mathrm{BTZ}} = 2\pi\sqrt{ch/6}$, and the Hawking–Page transition at $h = c/24$.
- The *subleading term* $2T/z$ encodes the state dependence of gravitational scattering. The braiding of Wilson lines (gravitational line operators) in the holomorphic-topological bulk is controlled by this term.
- The *pole structure* (cubic leading pole, simple subleading pole, no even poles) is the signature of a bosonic algebra extracted through the $d\log$ kernel: the OPE poles $z^{-4}, z^{-2}, z^{-1}$ each lose one order under the $d\log$ extraction, producing $r$-matrix terms $z^{-3}, z^{-1}, z^0$. The $z^0$ term (from the $\partial T$ pole at $z^{-1}$) is regular and does not contribute to the $r$-matrix, leaving only $z^{-3}$ and $z^{-1}$.

The contrast with Chern–Simons is maximal. In Chern–Simons gauge theory, the affine $r$-matrix $r^{\mathrm{aff}}(z) = \Omega/((k+h^\vee)z)$ is a simple pole and the coproduct is nontrivial: gluons split and the splitting is nontrivial. In gravity, the $r$-matrix has a cubic pole and the coproduct is trivial: gravitons braid but do not split. The Drinfeld–Sokolov reduction bridges the two: it takes the affine Casimir $\Omega/((k+h^\vee)z)$ to the gravitational $(c/2)/z^3 + 2T/z$ by homological perturbation through the Sugawara nonlinearity, and in the same stroke, the ghost-number obstruction kills the nontrivial coproduct.

| **Datum** | **Chern–Simons** | **Gravity** |
|---|---|---|
| boundary algebra | $V_k(\mathfrak{g})$ (affine KM) | $\mathrm{Vir}_c$ (Virasoro) |
| highest OPE pole | $z^{-2}$ (quadratic) | $z^{-4}$ (quartic) |
| $A_\infty$ tower | strict ($m_{k\geq 3} = 0$) | infinite ($m_k \neq 0$ all $k$) |
| shadow depth | 3 (class **L**) | $\infty$ (class **M**) |
| $r^{\mathrm{coll}}(z)$ | $\Omega/((k+h^\vee)z)$ (simple pole) | $(c/2)/z^3 + 2T/z$ (cubic pole) |
| $\Delta_z$ | nontrivial (gluons split) | primitive (gravitons braid) |
| $\kappa$ | $\dim\mathfrak{g}\,(k+h^\vee)/(2h^\vee)$ | $c/2$ |
| $\kappa + \kappa^!$ | 0 | 13 |

The entire table descends from a single datum: the number of $\lambda$-powers in the $\lambda$-bracket. Affine Kac–Moody has $\lambda^1$ at most (quadratic pole), yielding strict associativity, finite shadow depth, a simple-pole $r$-matrix, and a nontrivial coproduct. The Virasoro algebra has $\lambda^3$ (quartic pole), yielding the full infinite $A_\infty$ tower, infinite shadow depth, a cubic-pole $r$-matrix, and a primitive coproduct. The Drinfeld–Sokolov reduction is the functor that connects the two columns: it increases the $\lambda$-bracket order by two, trades the nontrivial coproduct for a ghost obstruction, and replaces the complementarity $\kappa + \kappa^! = 0$ with $\kappa + \kappa^! = 13$.

In one sentence: *gravity does not produce particles, only braids; the graviton is composite, and the ghost sector of its composite origin is the algebraic obstruction that kills the splitting.*